hafsteinne@hi.is
ORCID 0000-0001-5072-3678
Google Scholar · GitHub
haffi112.github.io

Hafsteinn Einarsson

01 Research Statement

I work on machine learning for settings that mainstream models serve poorly: small languages, specialised scientific measurements, and clinical text. My group builds the infrastructure that lets Icelandic and Faroese participate in modern NLP, from language models and translation resources to evaluation sets for gender bias, sentiment, and question answering. The same toolkit supports collaborations in marine and agricultural vision, automated clinical coding with the National Hospital, and human genetics at deCODE. I was trained in probabilistic combinatorics and models of cortical circuits. That background still shows in how the group works: explicit assumptions, small data budgets, and results we can check.

02 Current Positions

2026-present **Dean**, University of Iceland, Faculty of Industrial Engineering, Mechanical Engineering and Computer Science, Iceland
2024-present **Associate Professor in Computer Science**, University of Iceland, Faculty of Industrial Engineering, Mechanical Engineering and Computer Science, Iceland
2021-present **Principal Scientist**, Amgen deCODE, Iceland

03 Previous Positions

2024-present **Co-founder**, KatlaCode, Iceland
2020-2024 **Assistant Professor in Computer Science**, University of Iceland, Iceland
2019-2023 **Co-founder**, Prescriby, Iceland
2018-2020 **Data Scientist**, Íslandsbanki, Iceland
2017-2018 **Postdoctoral Researcher**, Institute of Neuroinformatics, ETH Zurich, Switzerland

04 Education

2017 **PhD in Computer Science**, ETH Zurich, Switzerland.
Fast Emergence of Neural Structures. Prof. Angelika Steger
2013 **MSc in Computer Science**, ETH Zurich, Switzerland.
2011 **BSc in Mathematics**, University of Iceland, Iceland.

05 Awards and Grants

2026 **Vísindi og velferð** — Grant for developing an app for science literacy, 700k ISK
2026 **Microsoft Lingua** — Grant for developing LLM red-teaming datasets for Icelandic (Microsoft), 50k USD
2024 **Science Award, University of Iceland** — Best overall science work and best in health science (for ICD-10 coding using Icelandic BERT models)
2023-present **TrustLLM** — EU-funded project on language models for Germanic languages (Horizon Europe)
2022 **Neologism platform for Icelandic** (University of Iceland), 4M ISK
2021 **Incentive Award, University of Iceland** — For QA models for Icelandic (supervised student project)
2020-2021 **Sproti and Vöxtur** — Awarded to Prescriby for digital product development (Icelandic Centre for Research (Rannís)), 90M ISK total
2020 **Digital literacy course development** (University of Iceland), 3M ISK
2017 **Early Postdoc Mobility Fellowship (declined)** (Swiss National Science Foundation)
2010 **Summer Undergraduate Research Fellowship (SURF)** (California Institute of Technology)

06 Supervision

PhD students (main supervisor) — 3

Annika Simonsen — TrustLLM: aligning language models for Germanic languages
Joao Rodrigo Da Silva Martins — PhD with the Marine and Freshwater Research Institute, Iceland
Steinunn Rut Friðriksdóttir — Mitigating prejudice in language models

PhD committee — 6

Amir Hamedpour
Bjarki Ármannsson
Isidora Glisic
Pétur Helgi Einarsson
Sindri Emmanúel Antonsson
Þór Sverrisson

Past PhD committees — 2

2026 **Emil Harðarson** — Artificial Intelligence in Regulated Domains: Lessons from Sleep Medicine
2025 **Andrea Rakel Sigurðardóttir** — Multispectral imaging

Master's theses completed — 14

2026 **Hörður Helgi Helgason** — Evaluating LLM performance on complex domain-specific classification tasks — automated evaluation of gold-plating in EU legal transposition
2026 **Yevhen Bilyi** — Predicting opioid tapering outcomes with machine learning: feature engineering and model comparison
2025 **Alexander Guðmundsson** — Evaluating Speech Technology Integration in Children's Reading Education
2025 **Joao Rodrigo Da Silva Martins** — Fish Species Classification in Controlled Underwater Environments Using Contrastive Language-Image Models → PhD with the Marine and Freshwater Research Institute, Iceland
2025 **Margrét Snorradóttir** — Creativity and ADHD: Exploring the Role of Hyperfocus and Divergent Thinking
2025 **Steinar Bragi Sigurðarson** — AI-assisted book recommendations for children
2024 **Haraldur Orri Hauksson** — ICD-10 classification for Icelandic
2023 **Annika Simonsen** — Improving Machine Translation for Faroese using ChatGPT-Generated Parallel Data → Now my PhD student
2023 **Atli Snær Ásmundsson** — Sentiment analysis with IceBERT for Icelandic blogs → PhD student at Reykjavík University
2023 **Hrafnhildur Hauksdóttir** — Heart failure pharmacotherapy after discharge at Landspítali
2023 **Þór Sverrisson** — Abstractive summarization for Icelandic → PhD at University of Iceland
2022 **Andrea Rakel Sigurðardóttir** — Few-shot nematode detection with multi-spectral imaging → PhD at University of Iceland
2022 **Kim Cosmo Ström** — Drawing Music
2021 **Vésteinn Snæbjarnarson** — QA method for Icelandic → PhD at University of Copenhagen and ETH Zurich

Current master's students — 1

Haukur Barri Símonarson

07 Teaching

Roughly 300+ students per academic year across these courses.

MSc **AI and Leadership** — Course in the Executive MBA programme at the University of Iceland on artificial intelligence for working leaders. Participants learn what current AI systems can and cannot do, where AI is useful in the everyday work of leadership (data, decision-making, employee performance and well-being, and the leader's own development), and what ethical questions come with putting AI to work inside a business. Classes include direct work with generative AI tools. Co-taught in 2025 with Sigrún Gunnarsdóttir, a servant-leadership researcher; taught alone in 2026. In Icelandic. *Taught:* 2025, 2026.

MSc — **The AI Lifecycle** — Master's course on the full lifecycle of a production AI system: gathering and preparing data, feature selection, training models, evaluating them, putting them into production, serving them as services, and monitoring and maintaining them once live. The semester is built around three larger team projects in which student teams compete to build the best working solution. *Taught:* Spring 2021, Spring 2022, Spring 2023, Spring 2024, Spring 2025, Spring 2026.

MSc — **Introduction to Deep Learning** — Master's introduction to deep neural networks, covering backpropagation, model training and regularisation, and architectures used for image, audio, and text. Generative, adversarial, and sequential models are part of the syllabus, with PyTorch as the implementation language throughout. Students present a paper or project of their own from the field as part of the assessment. Taught in English. *Taught:* Fall 2025.

MSc — **Large Language Models in an Icelandic Context** — Master's course at the School of Humanities asking how well large language models work in Icelandic and what their behaviour says about both the models and human language. Topics include methods for evaluating LLM linguistic competence in Icelandic, the risks that come with widespread chatbot use in smaller language communities (misinformation, bias and prejudice), and what cross-linguistic performance suggests about the way children and machines acquire language. Lead instructor: Iris Edda Nowenstein. Taught in Icelandic. *Taught:* Spring 2026.

MSc — **Natural Language Processing (Business Intelligence module)** — Two-week module on natural language processing taught inside the Business Intelligence master's course at the School of Business, where Helga Ingimundardóttir is the lead instructor. The module covers text preprocessing, embeddings, classification, named-entity recognition, and the use of large language models in analyst workflows. Taught in English. *Taught:* Spring 2022, Spring 2023, Spring 2024.

BSc / MSc — **Analysis of Algorithms** — Algorithm design and analysis, taught as a bridge course between the BSc and MSc programmes in computer science. Covers asymptotic complexity, algorithms for sorting and searching, graph algorithms and matrix computation, and the standard design strategies of divide-and-conquer, greedy methods, and dynamic programming. The course closes with the basics of intractability: NP-completeness, approximation algorithms, and randomised algorithms. Co-taught with Steinn Guðmundsson and Páll Melsted on a rotating basis. Taught in English. *Taught:* Spring 2021, Spring 2023, Spring 2025.

BSc — **Computers, Operating Systems and Digital Literacy** — Self-study BSc course developed in 2020 and modelled on MIT's "Missing Semester of your CS Education". Covers computer organisation, operating-systems basics, and the command-line and tooling skills students need across the rest of the degree. Designed for both Icelandic and international students to work through at their own pace.

BSc — **Discrete Mathematics for Computer Science** — First-year course covering the discrete-math foundations used throughout the rest of the computer-science degree: propositions and predicate logic, set theory and Boolean algebra, induction and recursion, counting and basic algorithm analysis, relations and their representations, trees and graphs, and the basics of strings, finite automata, and formal grammars. Enrolment runs 200–230 students per cohort, a mix of Icelandic and international students. *Taught:* Fall 2020, Fall 2021, Fall 2022, Fall 2023.

08 Open-source Projects

active — **FoQA** — The first dedicated question-answering dataset for Faroese, with extractive and generative variants. Built through translation and adaptation of existing QA benchmarks combined with native-speaker validation, and accompanied by baseline evaluations of multilingual and Faroese-tuned language models. Released alongside the RESOURCEFUL 2025 paper.

maintenance — **GameQA** — A gamified mobile-app platform for crowdsourcing multi-domain question-answering datasets. Demonstrated at EACL 2023 System Demonstrations and used to collect the RUQuAD-1 reading-comprehension dataset for Icelandic.

maintenance **Hotter and Colder** — An Icelandic sentiment corpus of blog comments annotated with nuanced labels for sentiment, emotions, toxicity, sarcasm, hate speech, sympathy, and related categories. Distributed via the University of Iceland, with the companion methodology paper at NoDaLiDa / Baltic-HLT 2025.

maintenance **IceBERT family** — A family of Icelandic BERT-style language models pre-trained on the Icelandic Crawled Corpus. Released alongside the LREC 2022 paper with Miðeind, the IceBERT checkpoints have become the foundation for much of the subsequent Icelandic NLP work in our group and elsewhere.

maintenance **Icelandic WinoGrande** — An Icelandic adaptation of the WinoGrande commonsense-reasoning benchmark, built to evaluate Icelandic language models on pronoun disambiguation and world-knowledge tasks. Released by Miðeind ehf.

active **MazeEval** — A benchmark for testing sequential decision-making in language models through navigation tasks in procedurally generated mazes. Single-author preprint with the benchmark released alongside the paper.

maintenance **MIM-GOLD-EL** — An entity-linking corpus for Icelandic built on top of the MIM-GOLD named-entity collection. Released with the University of Iceland and introduced at LREC 2022; the companion methodology paper appeared at the Dataset Creation for Lower-Resourced Languages workshop.

maintenance **NQiI - Natural Questions in Icelandic** — An open-domain question-answering dataset for Icelandic, adapted from the English Natural Questions benchmark. Released in two versions (v1.0 at LREC 2022 and an updated v1.1) and distributed through the University of Iceland for evaluating Icelandic QA systems.

maintenance **RUQuAD-1** — The Reykjavik University Question-Answering Dataset for Icelandic, a SQuAD-style reading-comprehension benchmark collected through the GameQA mobile platform. Distributed via the University of Iceland.

09 Talks and Outreach

2026 **Talk at the SAR forum**, SAR málþing, Reykjavik, Iceland `invited`

2026 **Results of the Icelandic AI usage survey**, AI usage survey presentation, Reykjavik, Iceland `outreach`

2026 **Research seminar**, Amgen deCODE, Reykjavik, Iceland `invited`

2026 **Eru risamállíkön verri á íslensku? (Are large language models worse at Icelandic?)**, UTmessan — AI Norðurljós track, Reykjavik, Iceland `invited`

2026 **AI in clinical practice**, Læknadagar (Icelandic Medical Days), Reykjavik, Iceland `invited`

2026 **AI for medical students**, Faculty of Medicine, University of Iceland, Reykjavik, Iceland `invited`

2025 **KatlaCode in the TrustLLM context**, TrustLLM / KatlaCode presentation `workshop`

2025 **Talk at Vísindavefurinn's 25th anniversary**, Vísindavefurinn 25th anniversary, Reykjavik, Iceland `outreach`

2025 **TrustLLM project presentation**, TrustLLM consortium meeting `workshop`

2025 **Prompt engineering for researchers**, Prompt engineering workshop, Reykjavik, Iceland `workshop`

2025 **AI in nursing**, Icelandic Nurses' Association (Félag hjúkrunarfræðinga), Reykjavik, Iceland `invited`

2025 **Data quality for AI**, SKÝ data quality conference, Reykjavik, Iceland `invited`

2025 **Teachers' attitudes and perceptions of AI usefulness in academia**, ICERI 2025 `workshop`

2025 **University of Iceland teachers' attitudes and experience with generative AI**, Menntakvika 2025, Reykjavik, Iceland `workshop`

2024 **AI and society in Iceland**, Þjóðarspegillinn (Mirror of the Nation), Icelandic social sciences conference, Reykjavik, Iceland `workshop`

2024 **AI and continuing education**, Símennt continuing education forum, Reykjavik, Iceland `outreach`

2024 **AI in health and life sciences**, Life and Health Sciences conference, Reykjavik, Iceland `invited`

2024 **AI in education research**, Menntakvika 2024, Reykjavik, Iceland `workshop`

2024 **AI in design**, NordDesign 2024 `invited`

2024 **Applications of BERT models towards automation of clinical coding in Icelandic**, NAACL 2024 Findings, Mexico City, Mexico `invited`

2024 **Clinical AI in Iceland**, Landspítali (Icelandic National University Hospital), Reykjavik, Iceland `invited`
2024 **AI in biology**, Faculty of Life and Environmental Sciences, University of Iceland, Reykjavik, Iceland `invited`
2024 **Talk on AI**, Veritas conference, Reykjavik, Iceland `invited`
2024 **AI in continuing education**, Continuing education conference (Framhaldsfræðsla), Reykjavik, Iceland `outreach`
2024 **Talk at the Science and Innovation Awards**, University of Iceland Science and Innovation Awards (Vísinda- og nýsköpunarverðlaun), Reykjavik, Iceland `invited`
2024 **AI in language teaching**, Association of German Teachers in Iceland, Reykjavik, Iceland `invited`
2024 **AI and Icelandic language teaching**, Association of Upper Secondary School Icelandic Teachers (Félag framhaldsskólakennara í íslensku), Reykjavik, Iceland `invited`
2024 **Mistök gervigreindar (Mistakes of artificial intelligence)**, UTmessan, Reykjavik, Iceland `invited`
2023 **AI in Nordic university administration**, NUAS (Nordic University Administrators) `invited`
2023 **AI in business**, MBA Programme, University of Iceland, Reykjavik, Iceland `invited`
2023 **AI in Nordic universities**, NORRIC `invited`
2023 **AI standards and governance**, Standards Council of Iceland (Staðlaráð), Reykjavik, Iceland `invited`
2023 **AI in adult education**, Rannís adult education forum, Reykjavik, Iceland `outreach`
2023 **AI for lifelong learners**, University of the Third Age (U3A), Reykjavik, Iceland `outreach`
2023 **AI in health sciences**, School of Health Sciences, University of Iceland, Reykjavik, Iceland `invited`
2023 **HPC for language modelling**, Icelandic HPC workshop, Reykjavik, Iceland `workshop`
2023 **AI and ethics**, UNESCO forum on AI and Ethics `invited`
2023 **AI and the legal profession**, Bifröst University, Faculty of Law, Bifröst, Iceland `invited`
2023 **Talk on AI**, Kvasir, Reykjavik, Iceland `invited`
2023 **AI in teaching**, Icelandic Teaching Academy conference (Kennsluakademían), Reykjavik, Iceland `invited`
2023 **AI in agricultural research**, Agricultural University of Iceland (Landbúnaðarháskólinn), Hvanneyri, Iceland `invited`
2023 **AI and psychology**, Faculty of Psychology, University of Iceland, Reykjavik, Iceland `invited`
2023 **AI and society**, Rotary Club of Seltjarnarnes, Seltjarnarnes, Iceland `outreach`
2023 **AI in finance and banking**, Central Bank of Iceland (Seðlabanki Íslands), Reykjavik, Iceland `invited`
2023 **AI and the legal profession**, Icelandic Bar Association (Lögfræðingafélag Íslands), Reykjavik, Iceland `invited`
2023 **AI in higher education**, University of Akureyri teaching conference, Akureyri, Iceland `invited`
2022 **Icelandic language resources for the ELRC**, ELRC workshop, Reykjavik, Iceland `workshop`
2022 **Computer vision for fish stock assessment**, Marine and Freshwater Research Institute of Iceland (Hafró), Reykjavik, Iceland `invited`
2022 **AI for the curious**, Vísindavaka (European Researchers' Night), Reykjavik, Iceland `outreach`
2022 **Icelandic resources at LREC**, LREC 2022, Marseille, France `workshop`
2022 **AI in engineering education**, Nordtek conference `invited`
2022 **UTmessan 2022**, UTmessan, Reykjavik, Iceland `invited`
2021 **HPC for language modelling**, Icelandic HPC workshop, Reykjavik, Iceland `workshop`
2021 **Guest lecture on language technology**, University of Iceland, Reykjavik, Iceland `invited`
2021 **Research seminar**, Amgen deCODE ML group, Reykjavik, Iceland `invited`
2021 **Public talk on AI**, Reykjavík City Library (Borgarbókasafn), Reykjavik, Iceland `outreach`
2021 **AI in education**, Menntaþing (Education forum), Reykjavik, Iceland `outreach`

2021 **Talk on language technology and AI**, Icelandic Language Technology Conference (Máltækniráðstefnan), Reykjavik, Iceland `invited`
2021 **UTmessan 2021**, UTmessan, Reykjavik, Iceland `invited`
2021 **Nationally broadcast interview on AI**, RÚV (Icelandic National Broadcaster), Reykjavik, Iceland `outreach`

010 Service

Reviewing (selected):

2025-present Icelandic Technology Development Fund (Tækniþróunarsjóður), research and innovation applications
2024-present ACL Rolling Review
2022-present LREC workshops
2023 Animals
2023 Sensors
2023 Applied Sciences
2021 NeurIPS workshops
2020-2021 Machine Learning (Springer)
2018 COSYNE
2018 Hippocampus
2017 SIROCCO

Roles:

2026-present **Board member**, Association of Computer Scientists in Iceland
2026 **Ad-hoc committee member**, Ethics committee, University of Iceland — single case
2025-present **Council member**, Professional Council, Icelandic Technology Development Fund (Tækniþróunarsjóður) — reviewing research and innovation applications
2025 **Leading author**, AI policy of the University of Iceland — chaired the preparatory committee; the final policy was drafted by three main authors. https://hi.is/gervigreindarstefna
2024-present **Vice-chair and founding member**, University of Iceland Centre for Artificial Intelligence — preparatory work from 2024; centre established 2025. https://ai.hi.is/en/about/
2024-2026 **Editorial board member**, Vísindavefurinn (Icelandic Science Web), University of Iceland
2021-2026 **Committee member**, Icelandic language committee, University of Iceland
2021-2024 **Committee member**, Teaching committee, School of Engineering and Natural Sciences, University of Iceland
2021-2023 **Committee member**, Equality committee, School of Engineering and Natural Sciences, University of Iceland
2020-2026 **President**, Association of Computer Scientists in Iceland
2019-2020 **Committee member**, SKÝ committee on data applications
2013-2016 **Board member, vice-president, president**, VMI (Association of scientific staff, CS Department, ETH Zurich)

011 Media Appearances

Selected interviews, opinion pieces and broadcast appearances on AI, language technology, and research. Titles are in Icelandic with English translations underneath. Linked entries open the original story.

2026-05-27 **Utanríkisráðherra óttast „Brexit-augnablik" í þjóðaratkvæðagreiðslu um ESB-aðild**
Iceland's foreign minister fears 'Brexit moment' in EU accession referendum · The Guardian
2026-05-03 **Gervigreindin gæti haft áhrif á ESB-atkvæðagreiðsluna**
AI could influence the EU referendum vote · Vísir
2026-04-15 **Kokkakonur, krakkfaraldur, Auddi Blö og gervigreindarmisskilningur**
Female chefs, a kratom epidemic, Auddi Blö, and AI misconceptions · RÚV (Morgunútvarpið) · radio
2026-04-08 **Gífurlegir hagsmunir í húfi ef tæknin kemst í rangar hendur**
Enormous interests at stake if the technology falls into the wrong hands · Vísir
2026-02-12 **Er skynsamlegt að fækka þeim sem læra íslensku?**
Is it sensible to reduce the number of people learning Icelandic? · Vísir · opinion

2025-12-16 **Hefur þú sokkið í gervigreindarfen?**
Have you sunk into the AI swamp? · Vísir (Bítið) · podcast

2025-11-23 **Stærsti þjófnaður allra tíma**
The largest theft of all time · RÚV

2025-11-01 **Getur gervigreindarsmjaður leitt til gervigreindargeðrofs?**
Can AI sycophancy lead to AI-induced psychosis? · Vísir

2025-09-26 **Dauðakaffi, hvað býr að baki gervigreind? og það nýjasta úr heimi konungsfjölskyldna**
Death cafés, what is behind AI?, and the latest from the royal families · RÚV (Samfélagið) · radio

2025-09-24 **Bein útsending: Gervigreind og vísindamiðlun**
Live broadcast: AI and science communication · Vísir · livestream

2025-09-14 **Veit um fjögur sjálfsvíg tengd gervigreind**
Aware of four suicides linked to AI · Vísir

2025-07-31 **Óheppilegt ef fölsk mynd varpar sök á saklausan mann**
Unfortunate if a fake image casts blame on an innocent person · Vísir

2025-07-28 **Styttist í háþróuð gervigreindarsvindl á íslensku**
Sophisticated AI scams in Icelandic are imminent · Vísir

2025-05-07 **Kapphlaup gervigreindarinnar, Sérsveitin, velsæld o.fl.**
The AI race, the special forces, well-being and more · RÚV (Morgunútvarpið) · radio

2025-02-02 **Vatnsnotkun gervigreindar áætluð allt að 6,6 milljarðar rúmmetra árið 2027**
AI water usage projected at up to 6.6 billion cubic metres by 2027 · RÚV

2024-10-03 **Af fordómum gervigreindar, Gísla Marteini og því sem þú getur gert til að hafa áhrif!**
On AI prejudice, Gísli Marteinn, and what you can do to make an impact · Vísir · opinion

2024-05-21 **Geta kóðað sjúkraskýrslur á íslensku**
Can code medical records in Icelandic · Morgunblaðið (mbl.is)

2024-02-25 **Sora-tæknin gerir myndbönd af öllu mögulegu mögulegt**
Sora technology makes videos of nearly anything possible · Vísir

2023-11-10 **200 milljóna styrkur til að þróa gervigreindarmállíkan**
200 million ISK grant to develop an AI language model · RÚV

2023-06-18 **Gervigreind, verðbólga og úkraínskur kjúklingur í Sprengisandi**
AI, inflation and Ukrainian chicken on Sprengisandur · Vísir

2023-05-02 **Verðum að grípa inn í áður en gervigreindin tekur völdin**
We must intervene before AI takes over · Vísir

012 Collaborations

Academic

- Marine and Freshwater Research Institute of Iceland (otolith age reading, fish species classification)
- TrustLLM partners (Faroese, German and other European groups)
- University of Copenhagen
- ETH Zurich
- Amgen deCODE
- Faculty of Food Science, University of Iceland
- Landspítali (Icelandic National Hospital) — clinical coding and heart-failure pharmacotherapy

Industry / practice

- VideometerLab — multispectral image analysis for nematode detection
- Horseday — horse gait detection from smartphone recordings

013 All Publications by Theme

NLP for Icelandic and Germanic languages

2026 Einarsson, H. (2026). **Cross-lingual mathematical reasoning in LLMs: Evaluating performance on Icelandic vs. English problems**. *Proceedings of the fourth workshop on resources and representations for under-resourced languages and domains (RESOURCEFUL-2026).*

2026 **Einarsson, H.**, Haraldsson, J. A., Derayat, Í. A., Lund, S. H., Magnússon, B. S. (2026). **Icelandic Math Eval: a competitive mathematics benchmark for large language models**. *Proceedings of the fifteenth language resources and evaluation conference (LREC 2026)*.

2026 Simonsen, A., Scalvini, B., Johannesen, U., Debess, I. N., **Einarsson, H.**, Snæbjarnarson, V. (2026). **BRAGD: Constrained multi-label POS tagging for Faroese**. *Proceedings of the fifteenth language resources and evaluation conference (LREC 2026)*.

2026 Simonsen, A., Debess, I. N., **Einarsson, H.** (2026). **Entity linking for Faroese using large language models with web search**. *Proceedings of the fourth workshop on resources and representations for under-resourced languages and domains (RESOURCEFUL-2026)*.

2026 Simonsen, A., Stenlund, M., Bungum, L., Volhardt, M. D. S., **Einarsson, H.** (2026). **Reformulate and create, don't translate: Creating natural prompts for underserved languages**. *Proceedings of the fifteenth language resources and evaluation conference (LREC 2026)*.

2026 Stenlund, M., Simonsen, A., Bungum, L., Ebert, J., Wang, J., Filatov, O., Myneni, H., Riedel, M., **Einarsson, H.** (2026). **Synthetic instruction generation for low-resource Nordic languages: Viability and limitations in LLM instruction-tuning**. *Proceedings of the fifteenth language resources and evaluation conference (LREC 2026)*.

2025 Arnardóttir, Þ., Einarsson, E. B., Juto, G. I., Helgason, T. P., **Einarsson, H.** (2025). **Wikiqa-is: Assisted benchmark generation and automated evaluation of icelandic cultural knowledge in llms**. *Proceedings of the third workshop on resources and representations for under-resourced languages and domains (RESOURCEFUL-2025)*.

2025 Chang, T. A., Arnett, C., Eldesokey, A., Sadallah, A., Kashar, A., Daud, A., Olanihun, A. G., Mohammed, A. L., Praise, A., Sharma, A. M., others (2025). **Global piqa: Evaluating physical commonsense reasoning across 100+ languages and cultures**. *arXiv preprint arXiv:2510.24081*.

2025 **Einarsson, H.** (2025). **MazeEval: a benchmark for testing sequential decision-making in language models**. *arXiv preprint arXiv:2507.20395*.

2025 **Einarsson, H.**, Simonsen, A., Nielsen, D. S. (2025). **Proceedings of the 1st workshop on nordic-baltic responsible evaluation and alignment of language models (NB-REAL 2025)**. *Proceedings of the 1st workshop on nordic-baltic responsible evaluation and alignment of language models (NB-REAL 2025)*.

2025 Friðriksdóttir, S. R., Nielsen, D. S., **Einarsson, H.** (2025). **Hotter and colder: a new approach to annotating sentiment, emotions, and bias in icelandic blog comments**. *Proceedings of the joint 25th nordic conference on computational linguistics and 11th baltic conference on human language technologies (NoDaLiDa/baltic-HLT 2025)*.

2025 Scalvini, B., Simonsen, A., Debess, I. N., **Einarsson, H.** (2025). **Prompt engineering enhances Faroese MT, but only humans can tell**. *Proceedings of the joint 25th nordic conference on computational linguistics and 11th baltic conference on human language technologies (NoDaLiDa/baltic-HLT 2025)*.

2025 Scalvini, B., Debess, I. N., Simonsen, A., **Einarsson, H.** (2025). **Rethinking low-resource MT: the surprising effectiveness of fine-tuned multilingual models in the LLM age**. *Proceedings of the joint 25th nordic conference on computational linguistics and 11th baltic conference on human language technologies (NoDaLiDa/baltic-HLT 2025)*.

2025 Schram, Á. B., Jóhannesdóttir, S., **Einarsson, H.**, Bjarnadóttir, M. K. (2025). **Áskoranir og tækifæri með notkun gervigreindar í háskólastarfi: Raddir kennara og hlutverk háskóla. Frumrannsókn á viðhorfi og reynslu kennara við HÍ**. *Menntakvika 2025: Ráðstefna í menntavísindum*.

2025 Schram, Á., Jóhannesdóttir, S., **Einarsson, H.**, Bjarnadóttir, M. K., Rogers, B. L. (2025). **Teachers'attitudes and perceptions of the usefulness of ai in academia: How should universities respond to a changing educational landscape?**. *ICERI2025 proceedings*.

2025 Simonsen, A., Nielsen, D. S., **Einarsson, H.** (2025). **FoQA: A Faroese question-answering dataset**. *Proceedings of the third workshop on resources and representations for under-resourced languages and domains (RESOURCEFUL-2025)*.

2024 Arnardóttir, Þ., Ingólfsdóttir, S. L., Símonarson, H. B., **Einarsson, H.**, Ingason, A. K., Þorsteinsson, V. (2024). **Beyond error categories: A contextual approach of evaluating emerging spell and grammar checkers**. *Proceedings of the 3rd annual meeting of the special interest group on under-resourced languages@ LREC-COLING 2024*.

2024 Arnardóttir, Þ., Ingólfsdóttir, S. L., Ingvarsson Juto, G., Símonarson, H. B., **Einarsson, H.**, Ingason, A. K., Þorsteinsson, V. (2024). **Icelandic GPT-SW3 for spell and grammar checking**. *Miðeind ehf.*

2024 Debess, I. N., Simonsen, A., **Einarsson, H.** (2024). **Good or bad news? Exploring GPT-4 for sentiment analysis for faroese on a public news corpora**. *Proceedings of the 2024 joint international conference on computational linguistics, language resources and evaluation (LREC-COLING 2024)*.

2024 **Einarsson, H.**, Lund, S. H., Jónsdóttir, A. H. (2024). **Application of ChatGPT for automated problem reframing across academic domains**. *Computers and Education: Artificial Intelligence*.

2024 **Einarsson, H.**, Friðriksdóttir, S. R., Arnardóttir, Þ. (2024). **Icelandic sentiment corpus (hotter and colder)**. *Háskóli Íslands*.

2024 Friðriksdóttir, S. R., **Einarsson, H.** (2024). **Gendered grammar or ingrained bias? exploring gender bias in Icelandic language models**. *Proceedings of the 2024 joint international conference on computational linguistics, language resources and evaluation (LREC-COLING 2024)*.

2024 Friðriksdóttir, S. R., Simonsen, A., Ásmundsson, A. S., Friðjónsdóttir, G. L., Ingason, A. K., Snæbjarnarson, V., **Einarsson, H.** (2024). **Ice and fire: Dataset on sentiment, emotions, toxicity, sarcasm, hate speech, sympathy and more in Icelandic blog comments**. *Proceedings of the fourth workshop on threat, aggression & cyberbullying@ LREC-COLING-2024*.

2024 Kristjánsdóttir, I. J., **Einarsson, H.**, Ingason, A. K. (2024). **Improving phrase structure parsing for icelandic**. .

2024 Simonsen, A., **Einarsson, H.** (2024). **A human perspective on GPT-4 translations: Analysing Faroese to English news and blog text translations**. *Proceedings of the 25th annual conference of the european association for machine translation (volume 1)*.

2023 Eggertsson, V. Á., Jóhannesson, B. G., **Einarsson, H.**, Loftsson, H. (2023). **Effective entity disambiguation in low-resource languages: a study of icelandic**. *2023 IEEE/WIC international conference on web intelligence and intelligent agent technology (WI-IAT)*.

2023 Friðriksdóttir, S., **Einarsson, H.** (2023). **The effect of data encoding on relation triplet identification**. *Proceedings of the 24th nordic conference on computational linguistics (NoDaLiDa)*.

2023 Skarphedinsson, N., Gudmundsson, B., Smari, S., Larusdottir, M. K., **Einarsson, H.**, Khan, A., Nyberg, E., Loftsson, H. (2023). **GameQA: Gamified mobile app platform for building multiple-domain question-answering datasets**. *Proceedings of the 17th conference of the european chapter of the association for computational linguistics: System demonstrations*.

2023 Sverrisson, Þ., **Einarsson, H.** (2023). **Abstractive text summarization for Icelandic**. *Proceedings of the 24th nordic conference on computational linguistics (NoDaLiDa)*.

2022 **Einarsson, H.**, Friðriksdóttir, S. R. (2022). **Bootstrapping icelandic knowledge graph data**. *ESSLLI*.

2022 Friðriksdóttir, S. R., **Einarsson, H.** (2022). **Fictionary-based games for language resource creation**. *Proceedings of the 2nd workshop on novel incentives in data collection from people: models, implementations, challenges and results within LREC 2022*.

2022 Friðriksdóttir, S. R., **Einarsson, H.** (2022). **Friðriksdóttir & einarsson (2022), bootstrapping icelandic knowledge graph data**. *University of Amsterdam/ Amsterdam University of Applied Sciences*.

2022 Friðriksdóttir, S. R., Daníelsson, H., Eggertsson, V., Jóhannesson, B. G., Loftsson, H., **Einarsson, H.** (2022). **MIM-GOLD-EL-entity linking corpus for Icelandic (22.01)**. *University of Iceland.*

2022 Skarphéðinsson, N., Guðmundsson, B., Smári, S., Lárusdóttir, M., **Einarsson, H.**, Kahn, A., Nyberg, E., Loftsson, H. (2022). **Reykjavik university question-answering dataset 1 (RUQuAD-1)-version 22.02**. *Reykjavik University.*

2022 Snæbjarnarson, V., **Einarsson, H.** (2022). **Cross-lingual QA as a stepping stone for monolingual open QA in Icelandic**. *Proceedings of the workshop on multilingual information access (MIA).*

2022 Snæbjarnarson, V., Símonarson, H. B., Ragnarsson, P. O., Ingólfsdóttir, S. L., Jónsson, H. P., Þorsteinsson, V., **Einarsson, H.** (2022). **Icelandic WinoGrande v1. 0**. *Miðeind ehf.*

2022 Snæbjarnarson, V., **Einarsson, H.** (2022). **Natural questions in icelandic**. *Proceedings of the thirteenth language resources and evaluation conference.*

2022 Snæbjarnarson, V., Einarsson, B. T. T., Auðunardóttir, I. I., Sæmundsson, U. I., Bjarnadóttir, H., Gunnarsson, H. V., **Einarsson, H.** (2022). **NQil v. 1.1**. *Háskóli Íslands.*

2022 Snæbjarnarson, V., Símonarson, H. B., Ragnarsson, P. O., Ingólfsdóttir, S. L., Jónsson, H., Þorsteinsson, V., **Einarsson, H.** (2022). **A warm start and a clean crawled corpus-a recipe for good language models**. *Proceedings of the thirteenth language resources and evaluation conference.*

2022 Steinunn Rut Friðriksdóttir, B. G. J., **Einarsson, H.** (2022). **Building an icelandic entity linking corpus**. *Proceedings of the workshop on dataset creation for lower-resourced languages within the 13th language resources and evaluation conference.*

Machine learning for the natural sciences

2025 Silva Martins, J. R., Bárðarson, H., Guðbrandsson, J., **Einarsson, H.** (2025). **Temporal aggregation of vision-language features for high-accuracy fish classification in automated monitoring**. *Ecological Informatics.*

2025 Hrólfsdóttir, A. Þ., Sigurðardóttir, A. R., Sveinsdóttir, H. I., Arason, S., Schultz, N., **Einarsson, H.**, Gudjónsdóttir, M. (2025). **Multispectral imaging techniques for evaluating physicochemical-, antioxidant, and sensory properties of Alaria esculenta and Saccharina latissima**. *Algal Research.*

2025 Sigurðardóttir, A. R., Sveinsdóttir, H. I., Schultz, N., **Einarsson, H.**, Gudjónsdóttir, M. (2025). **Multispectral imaging as a predictive tool for freshness of whole Atlantic cod: Compared with sensory, chemical and microbiological analysis**. *Applied Food Research.*

2024 Sigurðardóttir, A. R., Sveinsdóttir, H. I., Schultz, N., **Einarsson, H.**, Gudjónsdóttir, M. (2024). **Sequence segmentation of nematodes in atlantic cod with multispectral imaging data**. *Foods.*

2023 Davíðsson, H. B., Rees, T., Ólafsdóttir, M. R., **Einarsson, H.** (2023). **Efficient development of gait classification models for five-gaited horses based on mobile phone sensors**. *Animals.*

2023 Hamedpour, A., Sigurdsson, B. D., Peñuelas, J., Filella, I., **Einarsson, H.**, Latré, S., Stefánsson, T. (2023). **Analyzing the effect of the duration of soil warming on subarctic grasslands using high-resolution multispectral images**. *EGU general assembly conference abstracts.*

2023 Sigurðardóttir, A. R., Sverrisson, Þ., Jónsdóttir, A., Gudjonsdottir, M., Elvarsson, B. Þ., **Einarsson, H.** (2023). **Otolith age determination with a simple computer vision based few-shot learning method**. *Ecological Informatics.*

2022 Hamedpour, A., Sigurdsson, B. D., Penuelas, J., Filella, I., **Einarsson, H.**, Latré, S., Stefánsson, T. (2022). **Drought response analysis on a geothermally warmed Sub-Arctic grassland ecosystem using multispectral drone images**. *16th international circumpolar remote sensing symposium.*

Clinical AI and cardiology

2026 Ingimarsdóttir, I. J., Saldarriaga, C., Nielsen, N. C., **Einarsson, H.**, Goldfeder, S., Mewton, N., Barasa, A., Basic, C., Oerlemans, M. I., Niederseer, D., others (2026). **The impact of country income level on the management of heart failure with preserved ejection fraction: an international survey**. *ESC Heart Failure.*

2026 Myhre, P. L., Lindberg, F., Løgstrup, B. B., Ingimarsdottir, I. J., Vistnes, M., Schjødt, I., Ljungman, C., Savarese, G., Dridi, N. P., **Einarsson, H.**, others (2026). **Enhancing heart failure care through registries: lessons learned from Nordic Countries' implementation of guidelines**. *European Journal of Heart Failure.*

2026 Olafsdottir, S., Snaedal, S., **Einarsson, H.**, Gunnthorsdottir, I., Vishram-Nielsen, J., Ingimarsdottir, I. (2026). **Understanding non-prescription of guideline-directed medical therapy in acute heart failure and advanced kidney disease: a nation-wide study**. *European Journal of Heart Failure.*

2025 Bergmann, H. M., Úlfarsson, Æ. Ö., **Einarsson, H.**, Jóhannsdóttir, H. X., Gunnarsson, G. T., Hrafnkelsdóttir, T. J., Andersen, K., Ingimarsdóttir, I. J. (2025). **Unravelling ATTR cardiac amyloidosis in iceland: a nationwide epidemiological study**. *Cardiology.*

2025 Gudjonsson, P., Andersen, K., **Einarsson, H.**, Ingimarsdottir, I. (2025). **Survival impact of amiodarone therapy in heart failure: insights from the Icelandic Heart Failure registry**. *European Heart Journal.*

2025 Guidetti, F., Giraldo, C. I. S., Shchendrygina, A., Kida, K., Niederseer, D., Basic, C., Rainer, P. P., Załęska-Kocięcka, M., Ogola, E., Mohty, D., others (2025). **Differences in heart failure with preserved ejection fraction management between care providers: an international survey**. *European Journal of Heart Failure.*

2025 Gunnthorsdottir, I., Almarsdottir, A. B., Andersen, K., Gunnarsdottir, A. I., Svansdottir, E., **Einarsson, H.**, Ingimarsdottir, I. J. (2025). **Factors influencing medication adherence in heart failure patients—a survey among cardiac healthcare providers**. *Clinical Pharmacology & Therapeutics.*

2025 Ingimarsdottir, I. J., Vishram-Nielsen, J. K., **Einarsson, H.**, Goldfeder, S., Mewton, N., Barasa, A., Basic, C., Oerlemans, M. I., Niederseer, D., Shchendrygina, A., others (2025). **Diagnostic and therapeutic practice for HFpEF across continents and regions: An international survey**. *ESC Heart Failure.*

2024 Gunnthorsdottir, I., Andersen, K., Almarsdottir, A., Gunnarsdottir, A., Svansdottir, E., **Einarsson, H.**, Ingimarsdottir, I. (2024). **Factors believed by cardiac care professionals to influence medication adherence in heart failure patients**. *European journal of heart failure.*

2024 Hauksdottir, H., **Einarsson, H.**, Gunnthorsdottir, I., Gunnarsson, P., Ingimarsdottir, I. (2024). **Real-life management of heart failure: evaluation of medication optimisation in different phenotypes**. *European journal of heart failure.*

2024 Hauksson, H. O., **Einarsson, H.** (2024). **Applications of BERT models towards automation of clinical coding in icelandic**. *Findings of the association for computational linguistics: NAACL 2024.*

2024 Ingimarsdottir, I., Vishram-Nielsen, J., **Einarsson, H.**, Goldfeder, S., Mewton, N., Barasa, A., Basic, C., Oerlemans, M., Niederseer, D., Gustafsson, F., others (2024). **Diagnostic and therapeutic practice for heart failure with preserved ejection fraction around the world: an international survey**. *European journal of heart failure.*

2024 Ingimarsdóttir, I. J., Hansen, J. S., Bergmann, H. M., **Einarsson, H.** (2024). **The Icelandic Heart Failure Registry—A nationwide assessment tool for HF care and intervention in HF treatment**. *ESC Heart Failure.*

2024 Saldarriaga, C., Gracia, S. S. G., Mejia, M. I. P., Shchendrygina, A., Kida, K., Fauvel, C., Zaleska-Kociecka, M., Mapelli, M., **Einarsson, H.**, Guidetti, F., others (2024). **Diagnostic and therapeutic practice for Heart Failure with preserved ejection fraction around the world: An international survey**. *Current problems in cardiology.*

2024 Saldarriaga, C., Ines, C., Goldfeder, S., Palacio, M., Shchendrygina, A., Guidetti, F., Vishram-Nielsen, J., Ingimarsdottir, I., **Einarsson, H.**, Kida, K., others (2024). **Diagnostic and therapeutic practices for Heart Failure with preserved ejection fraction around the world: an international survey**. *European journal of heart failure.*

2023 Agnarsson, H., Hansen, J., Gunnarsson, D., **Einarsson, H.**, Gunnthorsdottir, I., Ingimarsdottir, I. (2023). **Epidemiology of acute heart failure admissions in Iceland: phenotype distribution and prognostic implications**. *European heart journal.*

2023 Gunnarsson, D. Á., Ingimarsdóttir, I. J., **Einarsson, H.**, Snædal, S. (2023). **Acute heart failure: causes and outcomes by stages of chronic kidney disease and heartfailure phenotypes**. *ESC heart failure 2023.*

Human genetics and computational biology

2025 Einarsson, G., Arnason, H. K., Gisladottir, R. S., Bjornsdottir, G., Thorgeirsson, T. E., Skuladottir, A., Walters, G. B., Saevarsdottir, S., Magnusson, M. K., Halldorsson, G. H., others (2025). **Variant in a taste receptor locus tied to changes in the use of insomnia medication**. *Biological Psychiatry Global Open Science.*

2024 Einarsson, G., Thorleifsson, G., Steinthorsdottir, V., Zink, F., Helgason, H., Olafsdottir, T., Rognvaldsson, S., Tragante, V., Ulfarsson, M. O., Sveinbjornsson, G., others (2024). **Sequence variants associated with BMI affect disease risk through BMI itself**. *Nature Communications.*

2024 Thorolfsdottir, R. B., Jonsdottir, A. B., Sveinbjornsson, G., Aegisdottir, H. M., Oddsson, A., Stefansson, O. A., Halldorsson, G. H., Saevarsdottir, S., Thorleifsson, G., Stefansdottir, L., others (2024). **Variants at the interleukin 1 gene locus and pericarditis**. *JAMA cardiology.*

Computational neuroscience

2018 Matheus Gauy, M., Lengler, J., **Einarsson, H.**, Meier, F., Weissenberger, F., Yanik, M. F., Steger, A. (2018). **A hippocampal model for behavioral time acquisition and fast bidirectional replay of spatio-temporal memory sequences**. *Frontiers in neuroscience.*

2018 Weissenberger, F., **Einarsson, H.**, Matheus Gauy, M., Meier, F., Mujika, A., Lengler, J., Steger, A. (2018). **On the origin of lognormal network synchrony in CA1**. *Hippocampus.*

2017 **Einarsson, H.** (2017). **Fast emergence of neural structures**. *ETH Zurich.*

2017 **Einarsson, H.**, Gauy, M. M., Lengler, J., Steger, A. (2017). **A model of fast hebbian spike latency normalization**. *Frontiers in computational neuroscience.*

2017 Weissenberger, F., Meier, F., Lengler, J., **Einarsson, H.**, Steger, A. (2017). **Long synfire chains emerge by spike-timing dependent plasticity modulated by population activity**. *International journal of neural systems.*

2014 **Einarsson, H.**, Lengler, J., Steger, A. (2014). **A high-capacity model for one shot association learning in the brain**. *Frontiers in computational neuroscience.*

Discrete mathematics and probabilistic combinatorics

2019 **Einarsson, H.**, Lengler, J., Mousset, F., Panagiotou, K., Steger, A. (2019). **Bootstrap percolation with inhibition**. *Random Structures & Algorithms.*

2018 **Einarsson, H.**, Lengler, J., Gauy, M. M., Meier, F., Mujika, A., Steger, A., Weissenberger, F. (2018). **The linear hidden subset problem for the (1+ 1) EA with scheduled and adaptive mutation rates**. *Proceedings of the genetic and evolutionary computation conference.*

2016 Lengler, J., Mousset, F., Panagiotou, K., Steger, A., **Einarsson, H.** (2016). **Connectivity thresholds for bounded size rules**. *Annals of Applied Probability.*